

Remote sensing introduction and application to passive microwave sea ice concentration

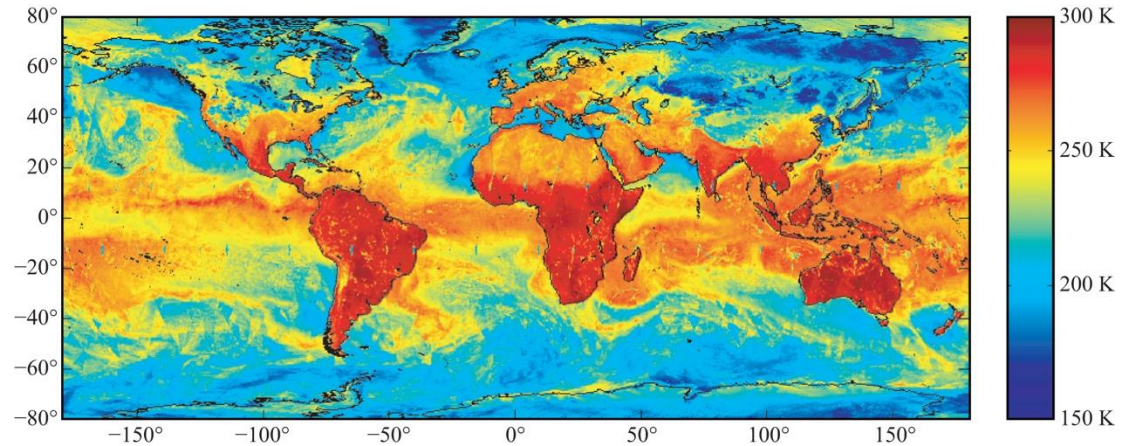
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May 11th 2026

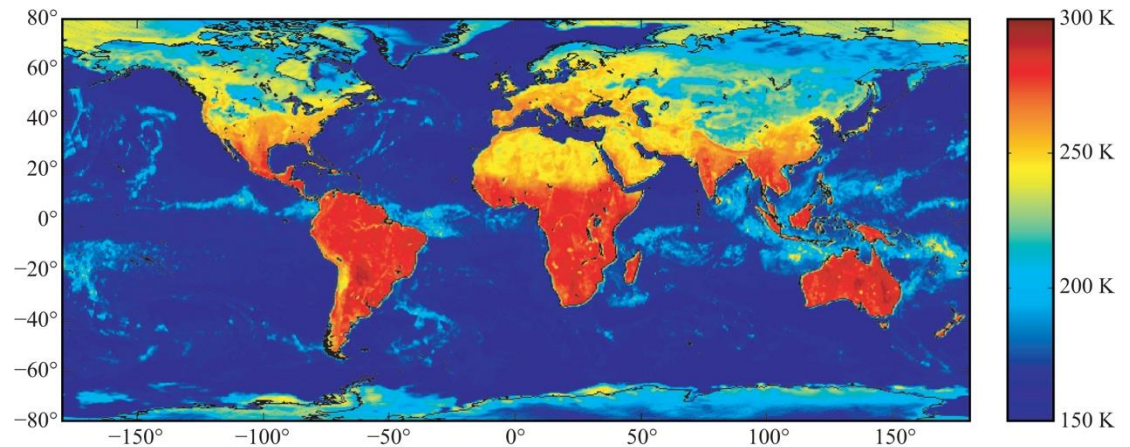
Remote sensing fundamentals

Remote sensing of the earth

- **Remote sensing** refers to the collection of information about an object without making physical contact with the object.
- Remote sensing instruments measure electromagnetic radiation
- Radiation is a type of energy transfer via electromagnetic fields



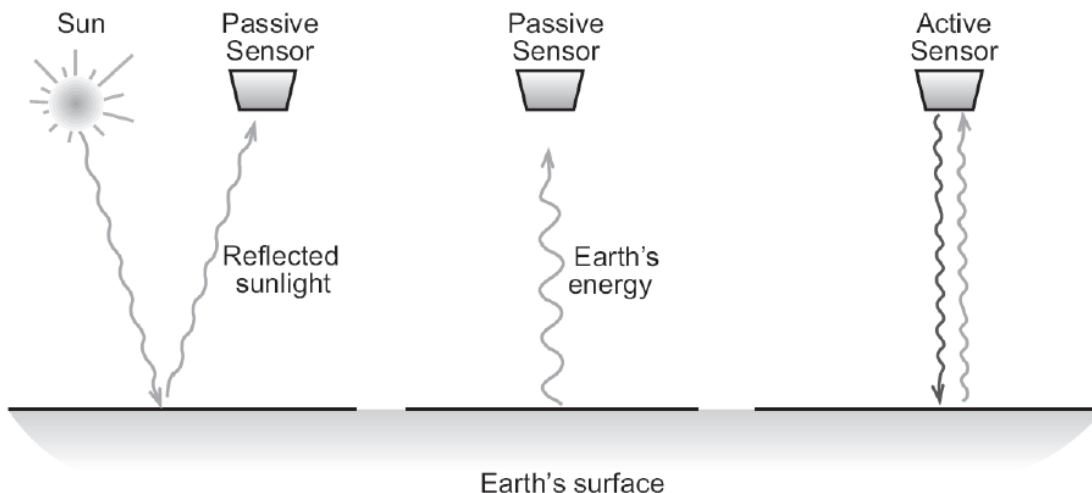
Brightness temperature, 89GHz



Brightness temperature, 36.5GHz

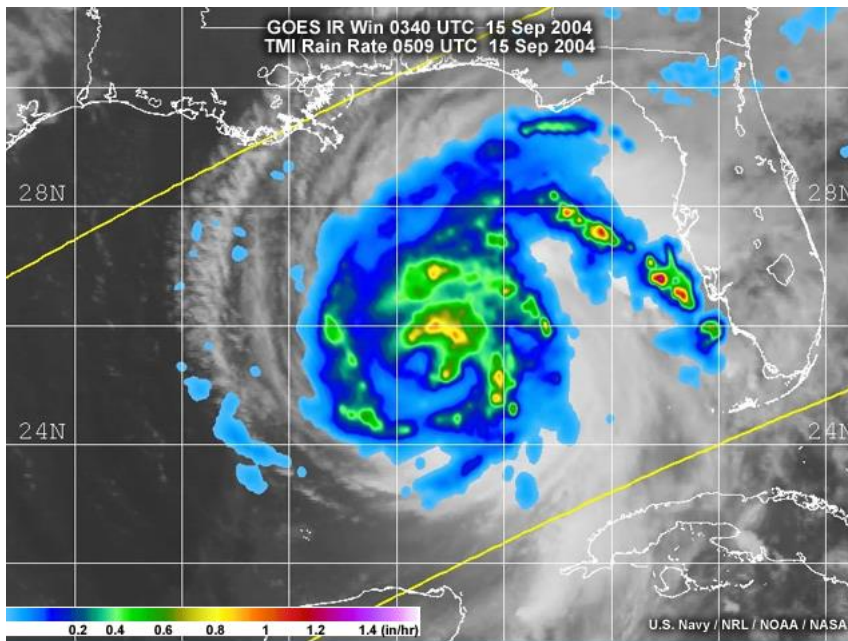
Passive vs active remote sensing

- Passive remote sensing systems measure naturally emitted radiation of the earth
- For passive systems we only know about the received signal
- For active systems a signal is both sent and received
- For active systems we know about the time between sent and received waves, and the properties of the sent and received signals

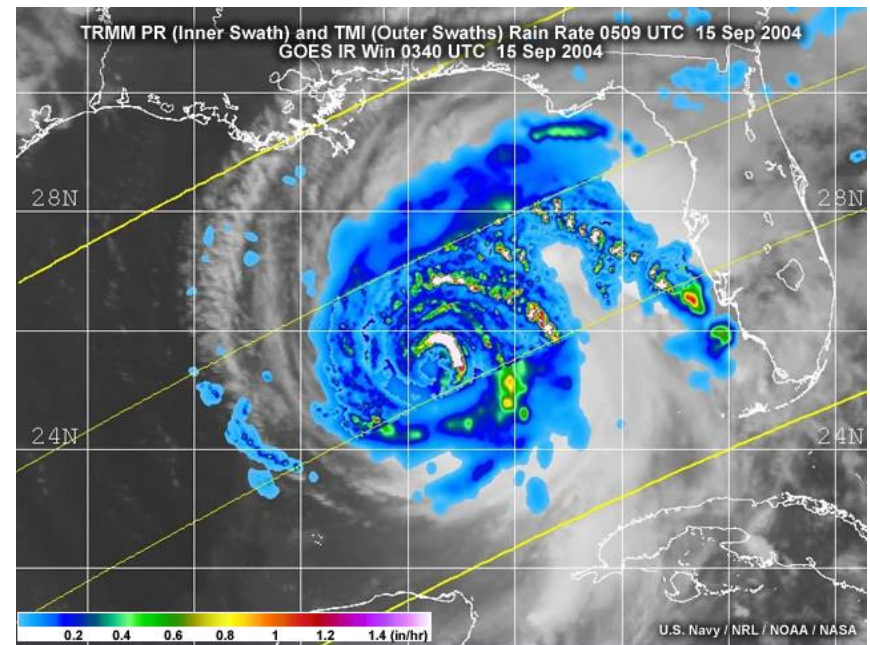


Passive vs active remote sensing: Rain rate example

- rain rates from the Tropical Rainfall Measuring Mission (TRMM), which has both an active and passive sensor on the same satellite
- Note the range of values and details are different for the two measurements



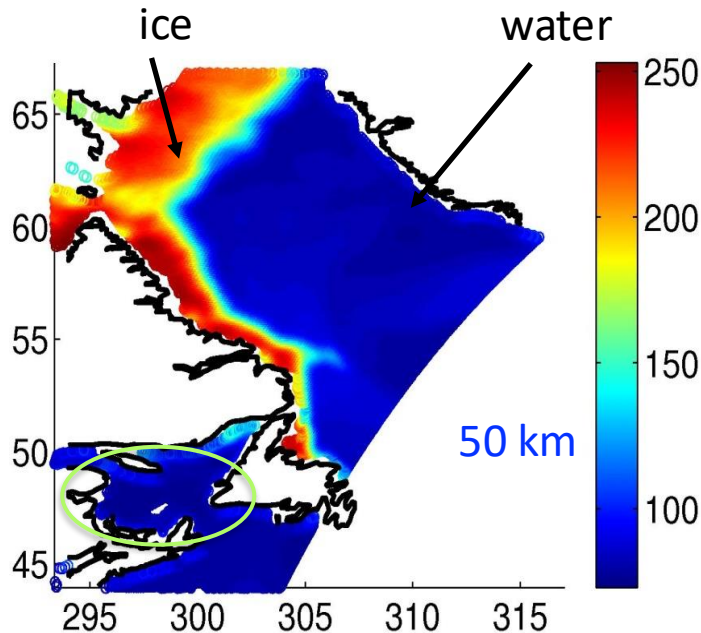
Passive sensor



Active sensor

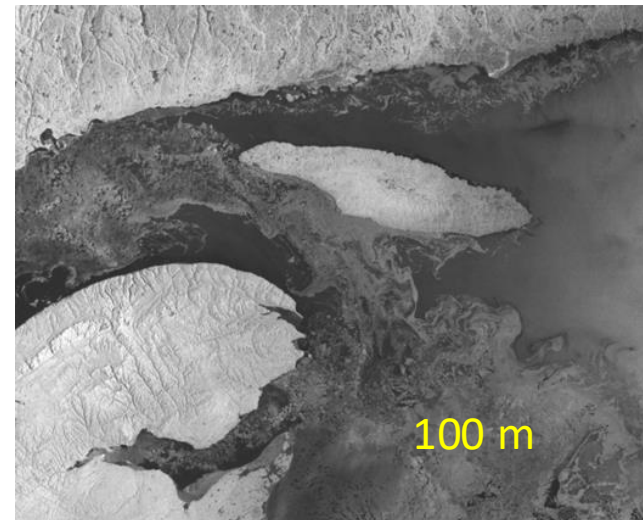
Passive vs active remote sensing: Sea ice example

Passive microwave (PM) - passive



- Brightness temperature: ice emits radiation more strongly than water
- Spatial resolution is coarse
- This is a measurement of low frequency (6.9 GHz) emission

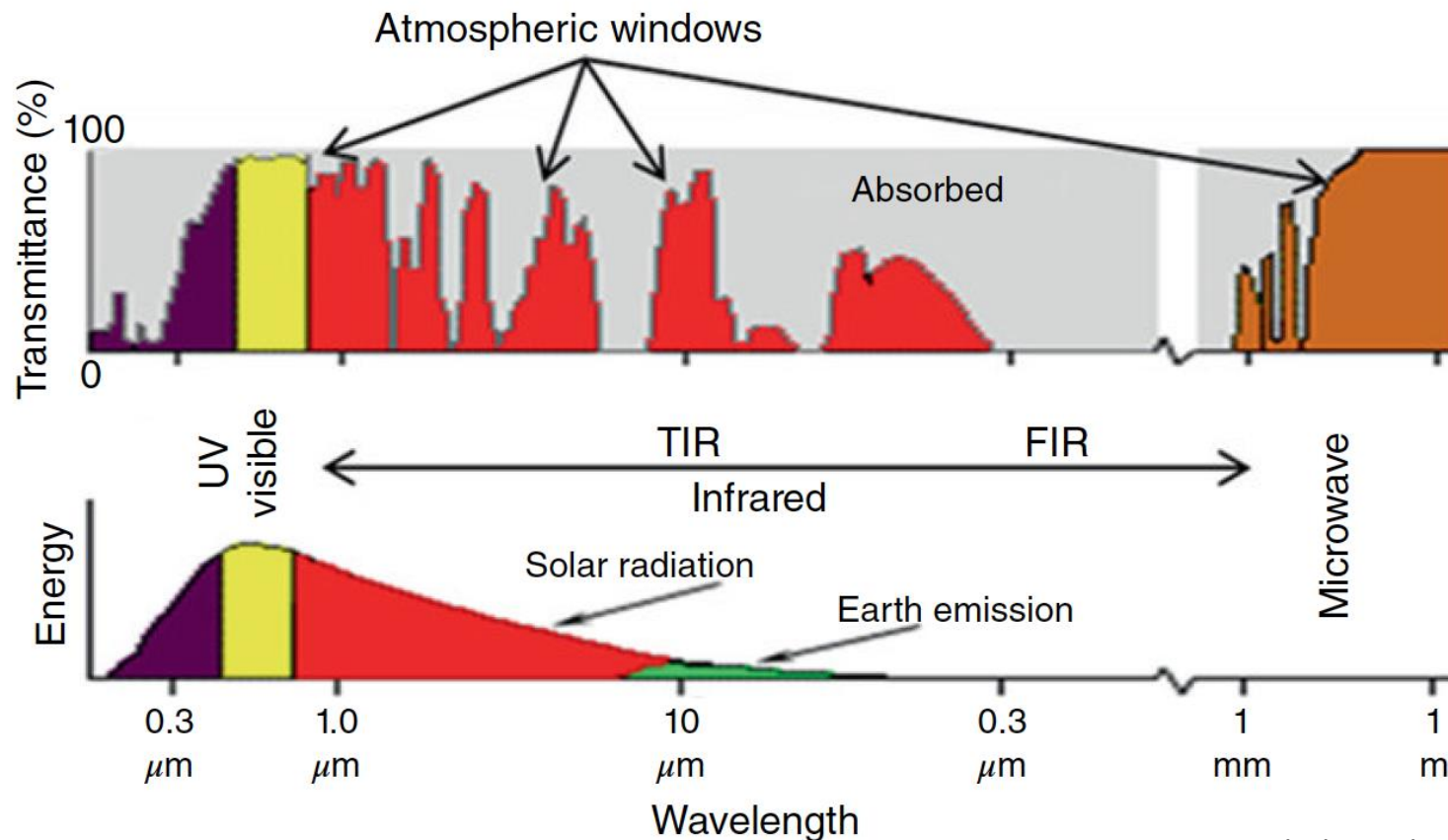
Synthetic aperture radar (SAR) - active



- Backscatter in the Gulf of St Lawrence
- Complex patterns for sea ice
- Tonal variability (bright/dark regions)
- High spatial resolution

Electromagnetic spectrum

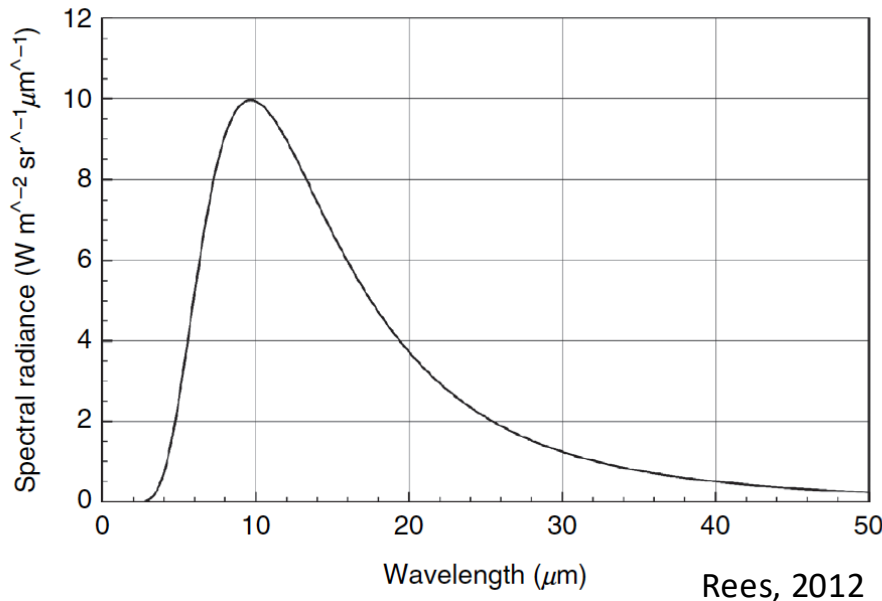
- Low frequency microwave in an atmospheric window -> little absorption by the atmosphere, surface of the earth can be relatively directly imaged



Planck's Law

- Previous slide we note the bottom panel, the sun and earth emission have a similar shape -> this is governed by Planck's law
- for a blackbody at a given temperature, radiation is emitted over a range of frequencies (or wavelengths) as described by Planck's law

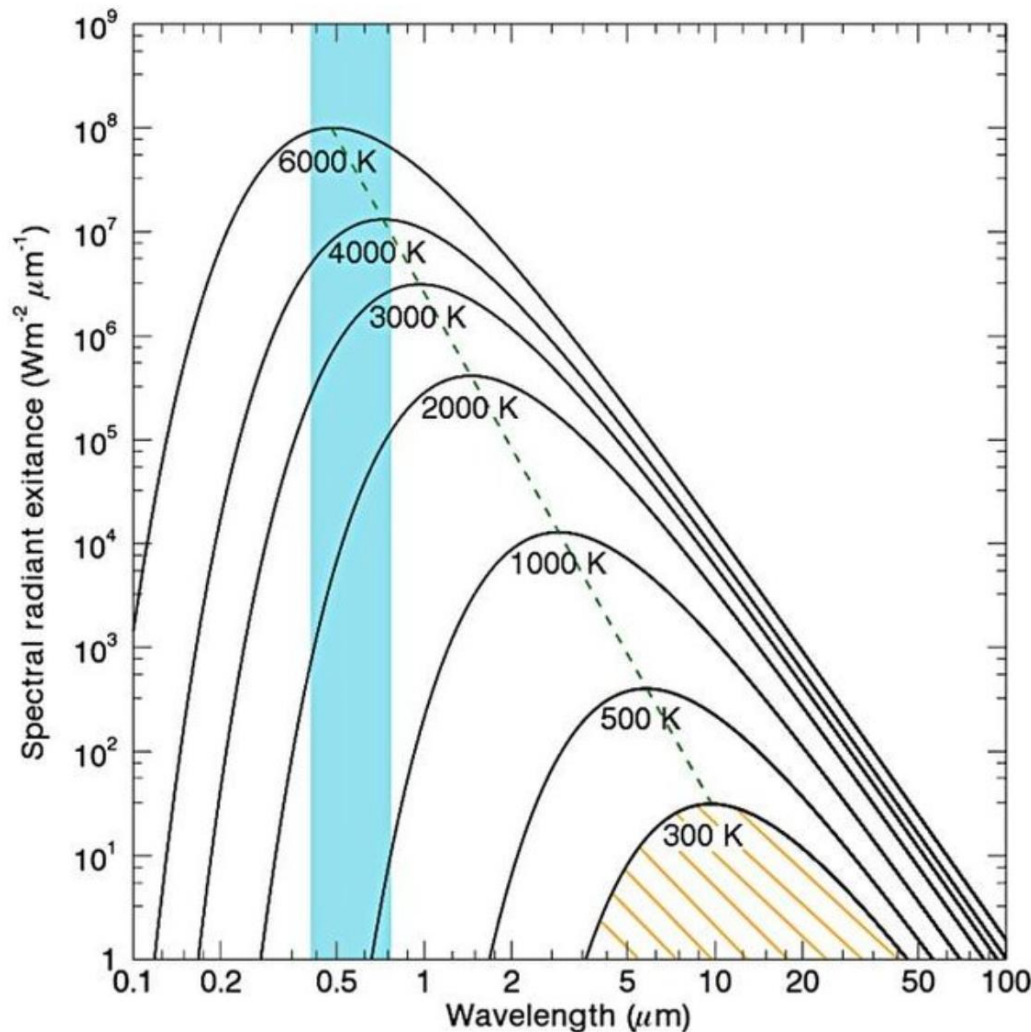
Spectral radiance of object at 300K



$$R(\lambda) = \frac{2hc^2}{\lambda^5} \left[\frac{1}{e^{hc/\lambda kT} - 1} \right]$$

$R(\lambda)$ Radiance [W/m²Sr]
 λ wavelength of radiation
 c speed of light
 k is Kirchoff's constant
 h Planck's constant
 T temperature of object

Planck's Law for various temperatures



- Sun ($\sim 6000\text{K}$) peaks in the visible range
- Earth ($\sim 300\text{K}$) peaks in the infrared range
- Wein's law $\lambda_{max} \sim 1/T$
- Wavelength of maximum radiance decreases with increasing temperature

Brightness temperature

- for a blackbody: measure R_B for a given λ -> obtain the temperature of the emitting body

$$R_B = \frac{2hc^2}{\lambda^5} \left[\frac{1}{e^{hc/\lambda kT} - 1} \right]$$

- a real body (or grey body) will emit a lower spectral radiance, R_G
- for a graybody: measure R_G for a given λ -> obtain the brightness temperature (T_B) of emitting body
- the ratio between the radiance emitted by a graybody to a blackbody is called the emissivity

$$\epsilon = \frac{R_G}{R}$$

Emissivity

- the ratio between the radiance emitted by a graybody to a blackbody is called the emissivity

$$\epsilon = \frac{R_G}{R}$$

- for the infrared part of the EM spectrum, emissivity is relatively constant across surface types
- it is generally around 0.9 to 0.99 (Lillesand, 2008)

Material	Average emissivity (infrared)
Clear water	0.98-0.99
Green vegetation	0.96-0.99
Dry soil	0.92-0.96

- for the microwaves it is much more variable – why this difference?

Kirchhoff's Law

- For a body in thermal equilibrium the amount of radiation absorbed is equal to that emitted

$$\alpha = \varepsilon$$

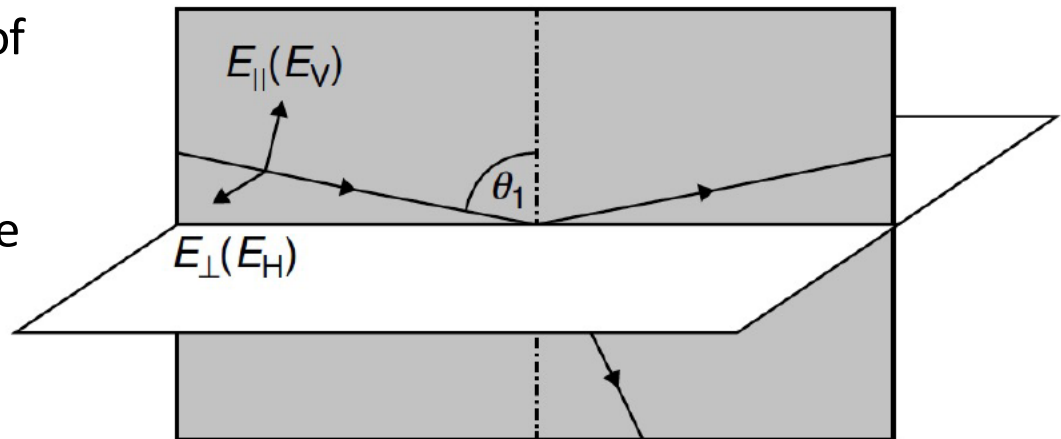
- Where α is the absorptivity and ε is the emissivity
- Materials that absorb well have a high emissivity
- In the infrared part of the EM spectrum, materials in the natural environment do absorb well, while in the microwave part of the EM spectrum
- To understand these, and other, differences better, we need to look at the EM waves and how they interact with media

Electromagnetic waves

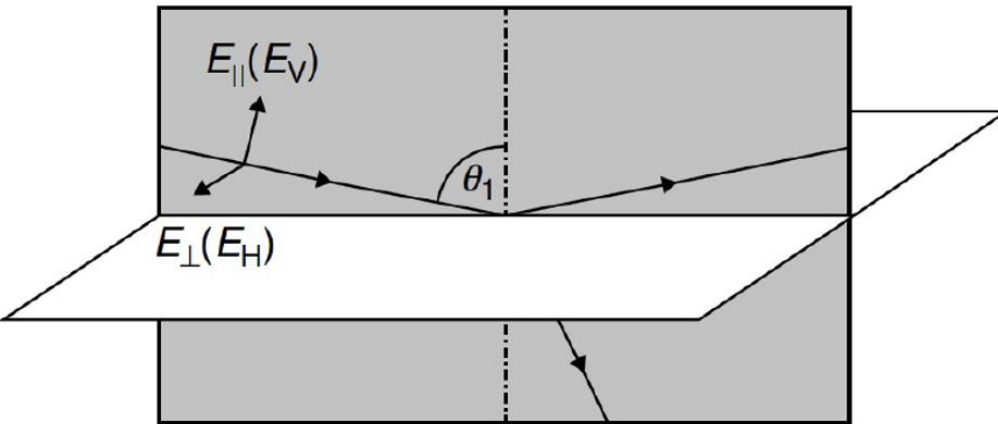
- EM waves have coupled electric and magnetic fields that are perpendicular to each other in free space (only one needs to be specified)
- The electric field vector can be written as having two orthogonal components, one horizontal and one vertical

$$\mathbf{E} = a_h e^{i\delta_h} \hat{\mathbf{h}} + a_v e^{i\delta_v} \hat{\mathbf{v}}$$

- the horizontal component is perpendicular to the plane of incidence
- the vertical component is within the plane of incidence
- the plane of incidence contains the normal to the surface and the incoming propagation vector



Fresnel's Equations - summary



Rees, 2012

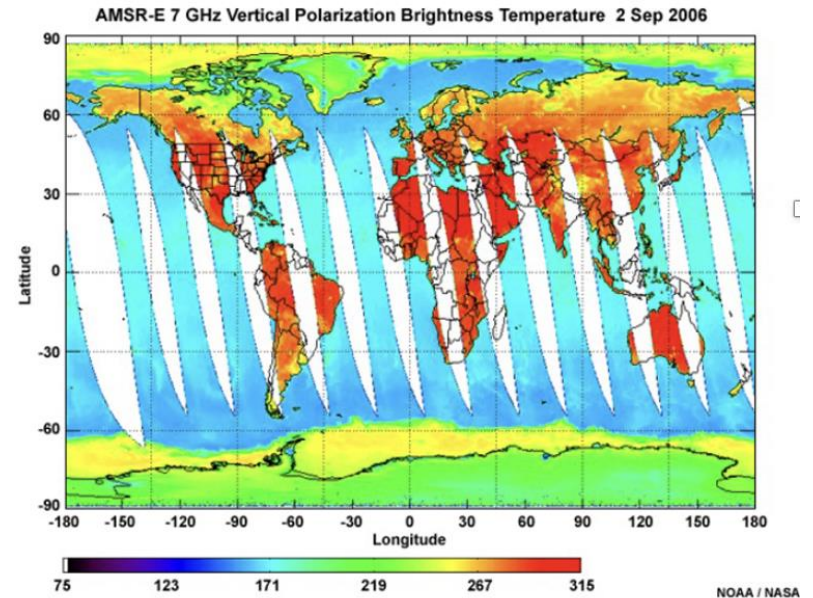
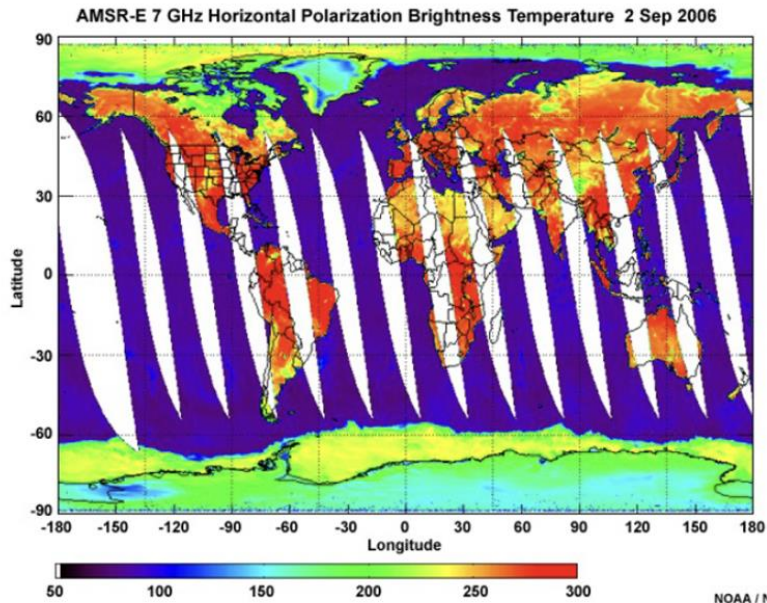
- The idea with Fresnel's equations is to obtain a relationship between the reflectance of the reflected to incident waves (or the power ratio)
- To do this you apply continuity conditions to the electric and magnetic fields at the boundary between the two media – noting that the magnetic field vector is perpendicular to the electric field vector
- You can do this for the horizontally polarized wave and then for the vertically polarized wave, and get a reflectance for each
- Full details are in Ulaby

Horizontal vs vertical polarizations

- Looking at the equations for the reflection coefficients, we can go a little farther if we assume that the first medium is air and recall that the relationship between the refractive indices of the two media are related by Snell's Law
- Brewster angle water... Rees page 55
- Note that Fresnel's equations only account for reflection, not scattering, which is important for smaller wavelengths
- This, in combination with other factors (like emissivity and sensor viewing geometry), means polarization is not usually measured for optical or infrared spaceborne sensors, but it is very important for the microwaves

Polarization: Sensitivity to roughness

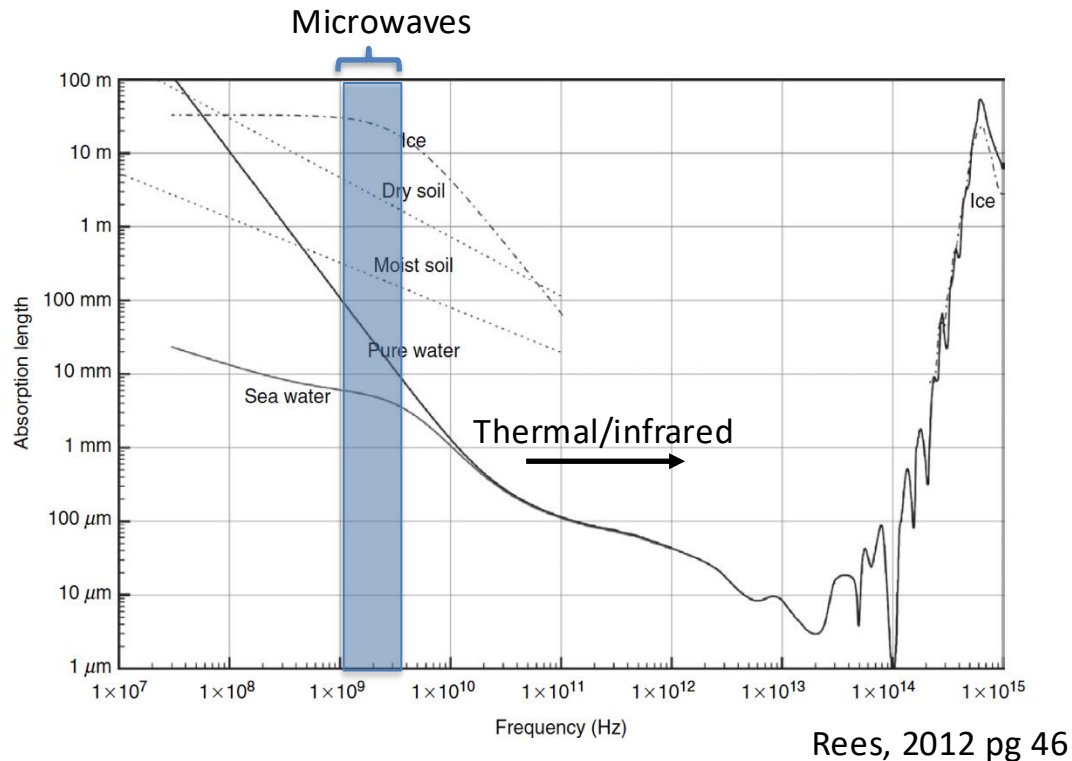
Polarization: Sensitivity to materials



- Note the brightness temperature are very different for water for the horizontal vs vertical polarizations

Material permittivity

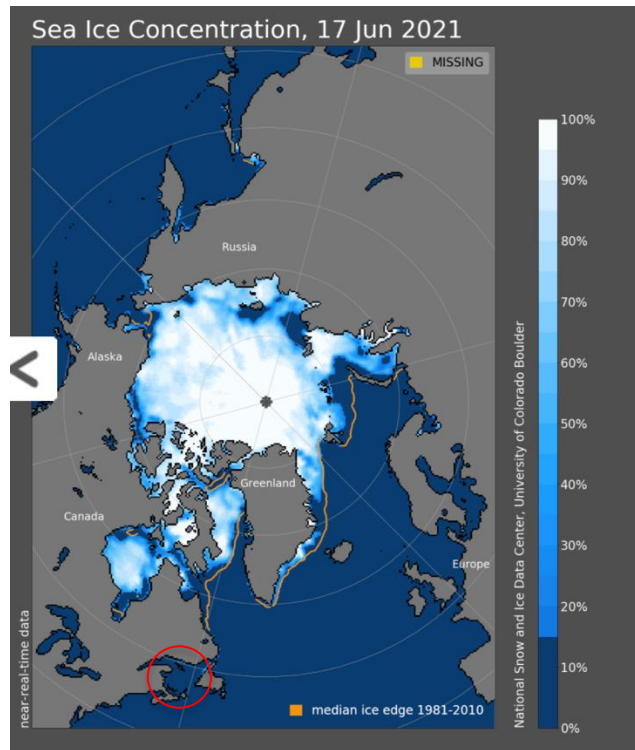
Absorption length or emitting layer depth



- Important because – different sensors will measure the surface at a different depth (sea surface temperature from infrared and microwave are NOT the same)
- For sea ice (not shown) the emitting layer depth is highly variable -> it is difficult to know precisely from what layer the emission is coming from

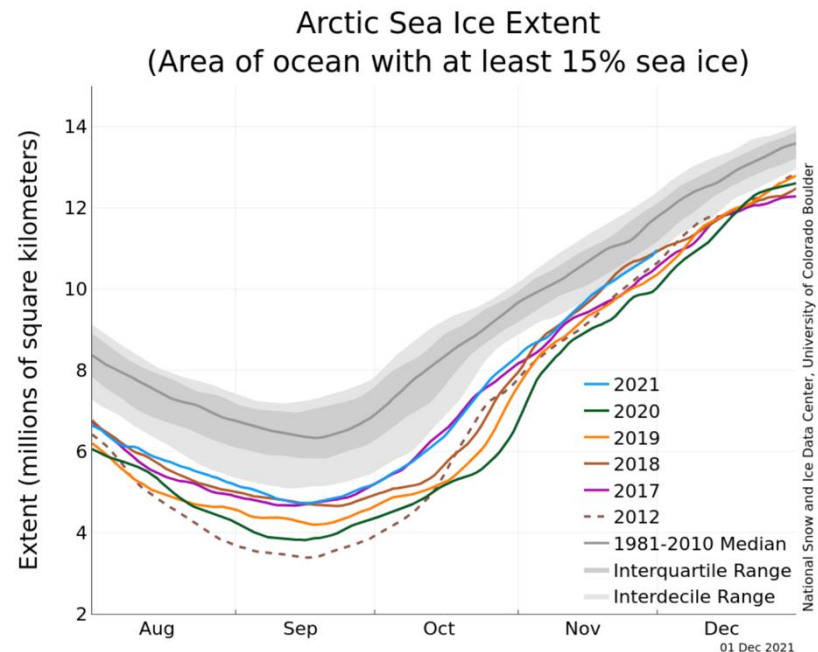
Passive Microwave: Sea ice Concentration

Passive microwave data: the workhorse of sea ice estimation for climate monitoring



NSIDC

- 40+ year data record crucial for monitoring
- Limitations due to coarse spatial resolution and weather
- Inability to differentiate between thin ice and low concentration regions. -> heat transfer!



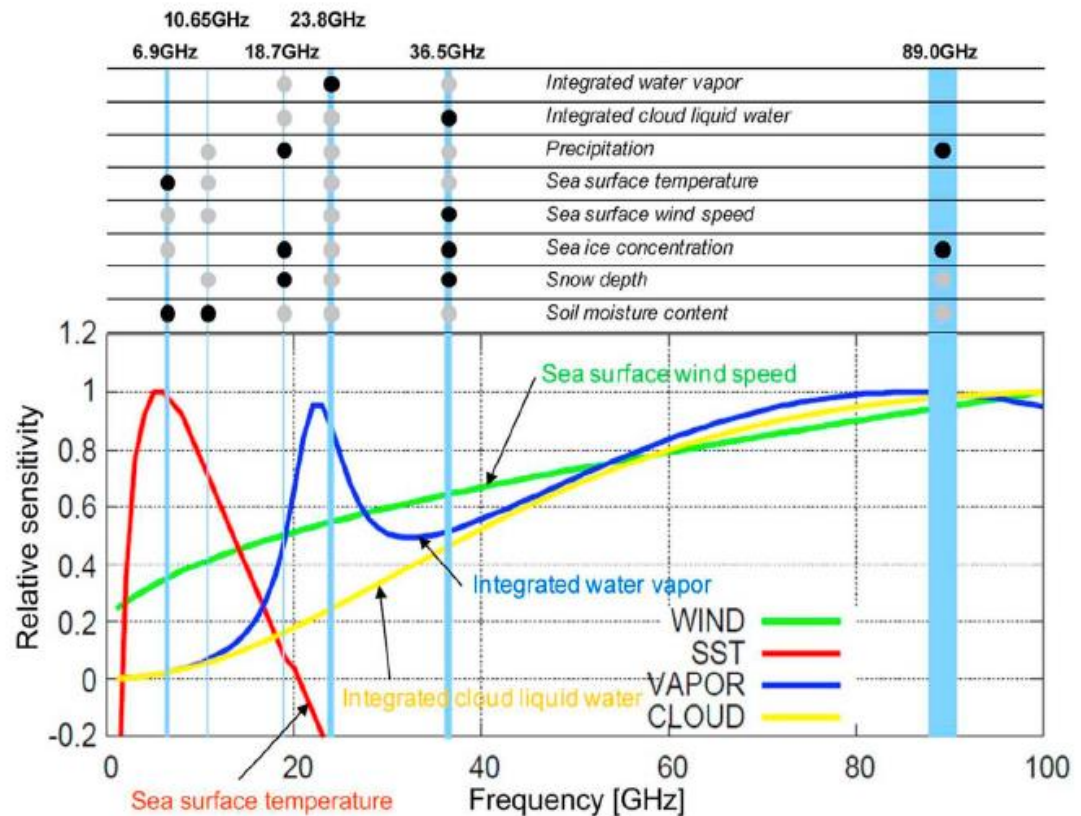
NSIDC

Passive microwave data

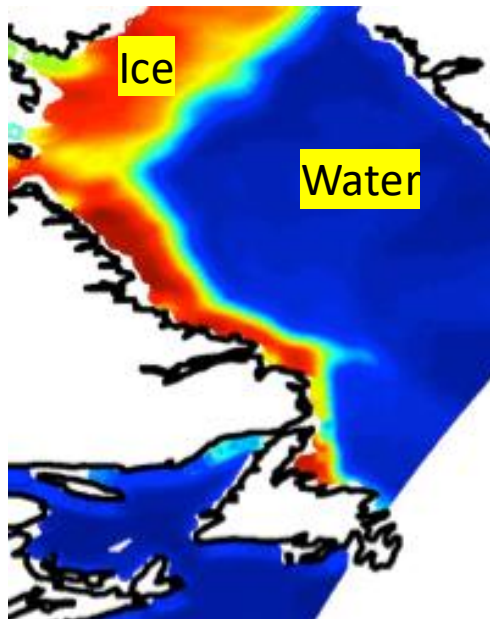
- passive microwave data provide near continuous coverage of the **earth's surface**
 - sunlight is not required
 - cloud cover does not block emission
 - data record extends over 40 years

Can be used to retrieve

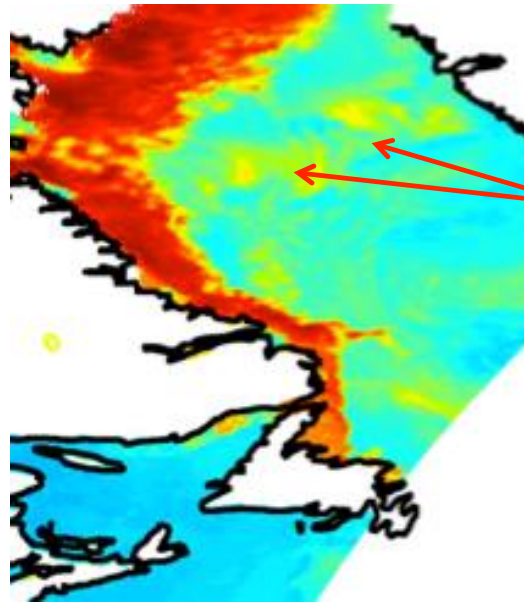
- surface temperature
- windspeed
- salinity
- water vapor
- cloud liquid water
- sea ice concentration



Brightness temperature: Frequencies

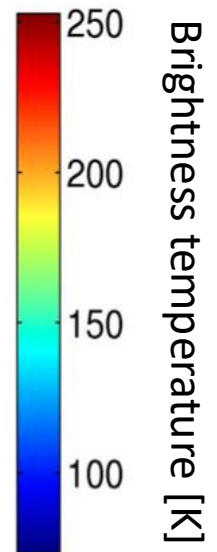


6.9GHz
coarse resolution



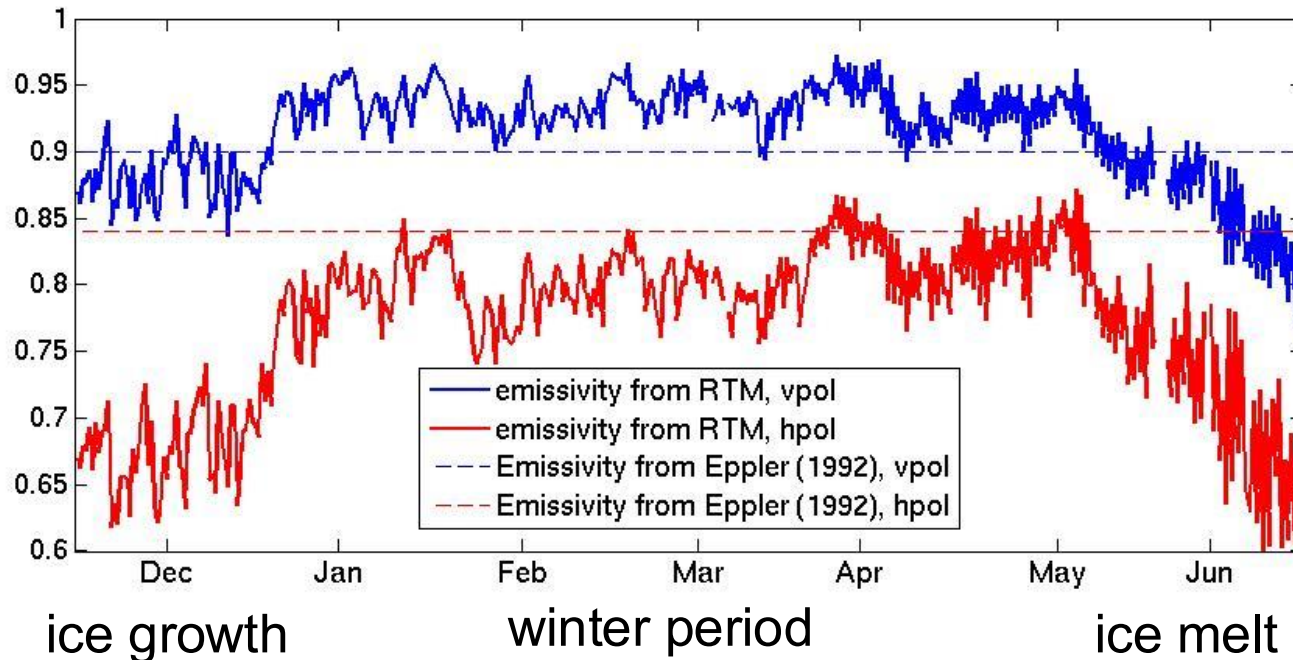
36.5GHz
higher resolution

Weather''
Effects!'



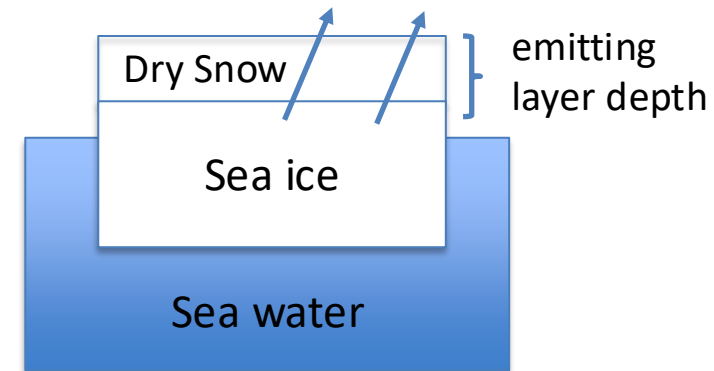
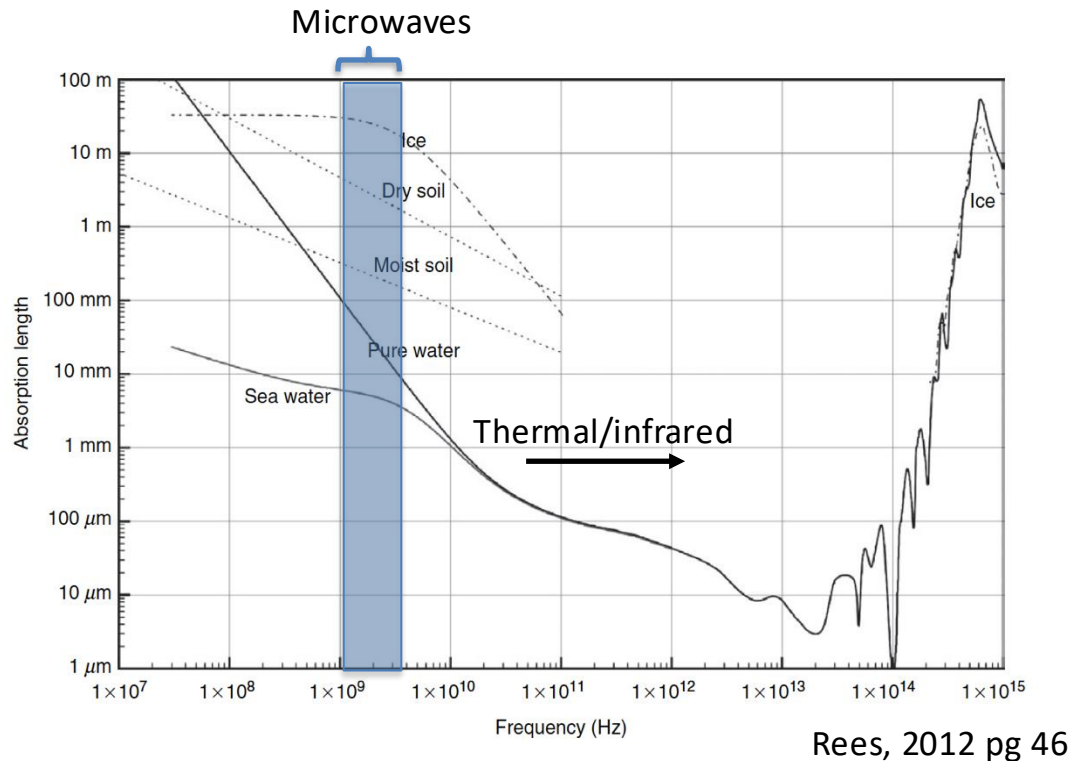
- We have more notable weather effects at the higher frequency (smaller wavelength)
- These weather effects increase the brightness temperature over water and generate a signal that is easily confused with sea ice of intermediate ice concentration

Emissivity: Microwaves for sea ice



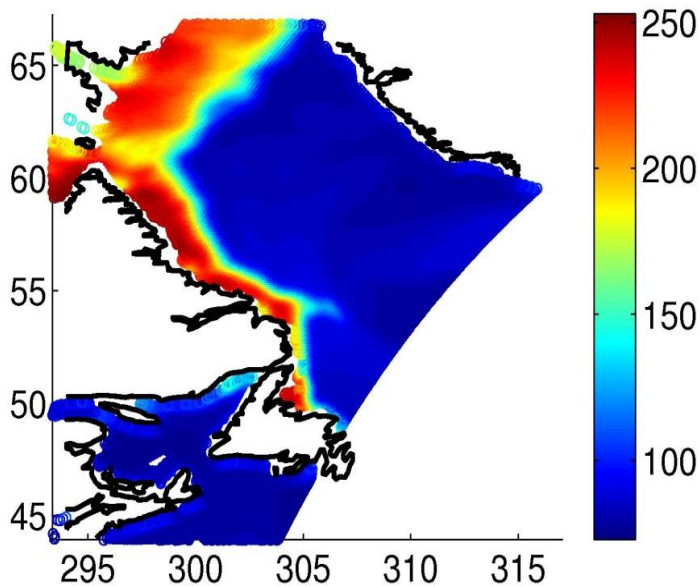
- There are big differences across seasons
- During freeze-up the ice is thin, and salty
- During ice melt there can be ponding on the surface
- The surface where the emission is coming from also depends on the sea ice and snow properties

Absorption length or emitting layer depth



- Important because – different sensors will measure the surface at a different depth (sea surface temperature from infrared and microwave are NOT the same)
- For sea ice (not shown) the emitting layer depth is highly variable -> it is difficult to know precisely from what layer the emission is coming from

Sea ice concentration from brightness temperature data: Single channel



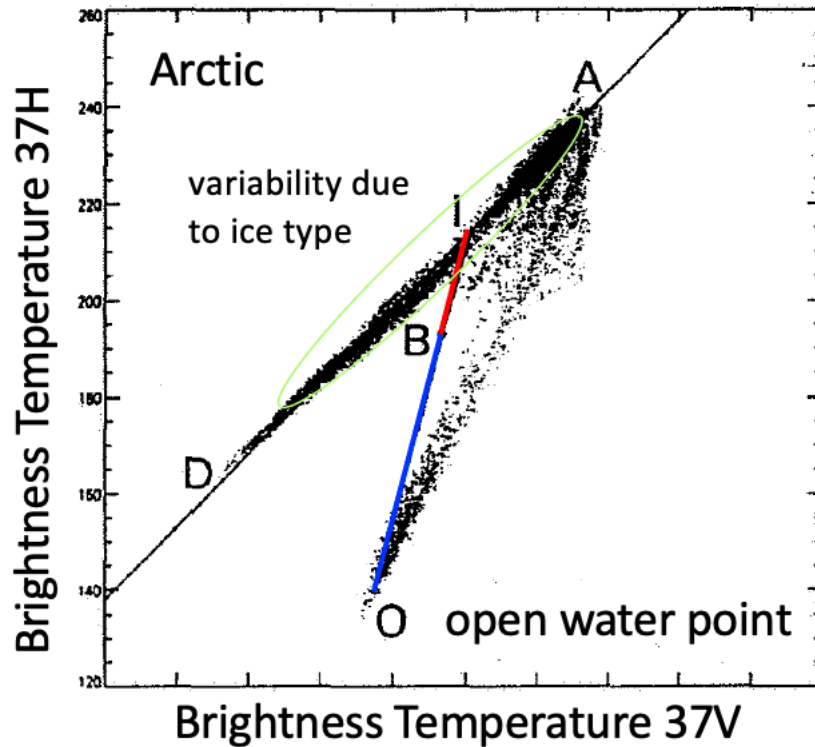
- there is a characteristic brightness temperature for ice (T_I)
- there is a characteristic brightness temperature for water (T_w)
- Represent measurement (BT) with linear mixture model

$$BT = CT_I + (1 - C)T_w$$

- Rearrange for ice concentration

$$C = \frac{BT - T_w}{T_I - T_w}$$

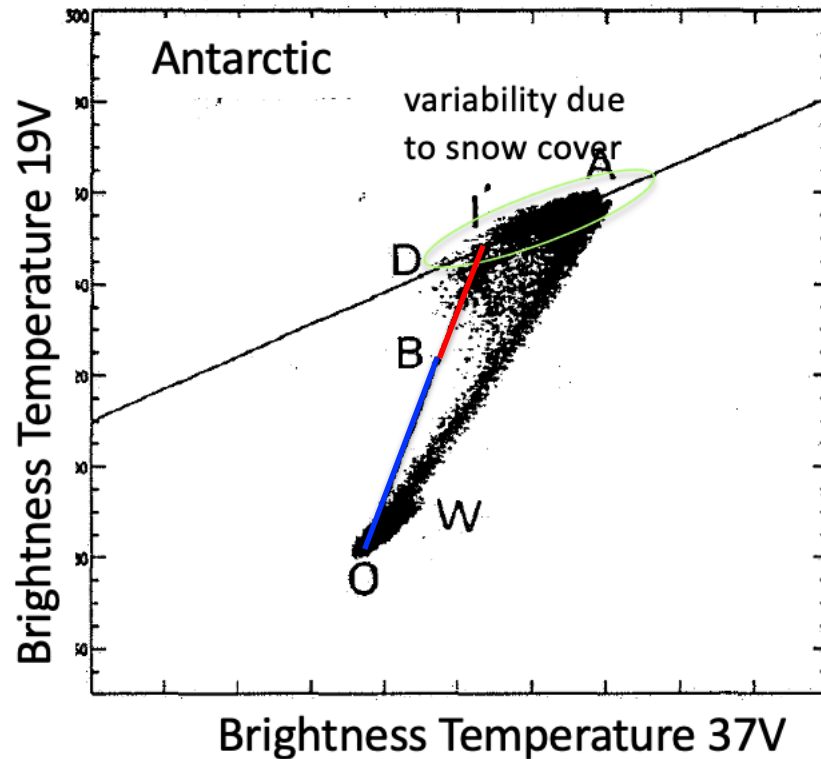
Multi-channel methods: Arctic



$$SIC \approx \frac{OB}{IO}$$

- Recall $BT = \epsilon T_s$
- Emissivity varies widely between new ice and older (multi-year ice)
- Ice tie point, T_I is variable (line AD)

Multi-channel methods: Antarctic



$$SIC \approx \frac{OB}{OI}$$

- There is little multi-year ice in Antarctica
- Ice tie point, T_I is variable (line AD) due to snow properties (thickness, wetness and layering)

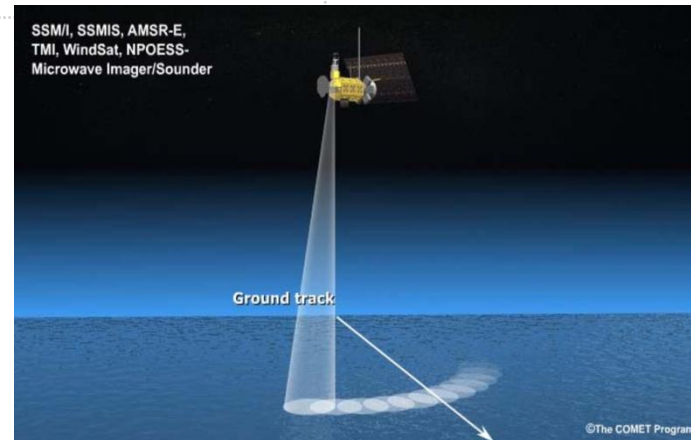
Passive microwave data

Passive microwave data page

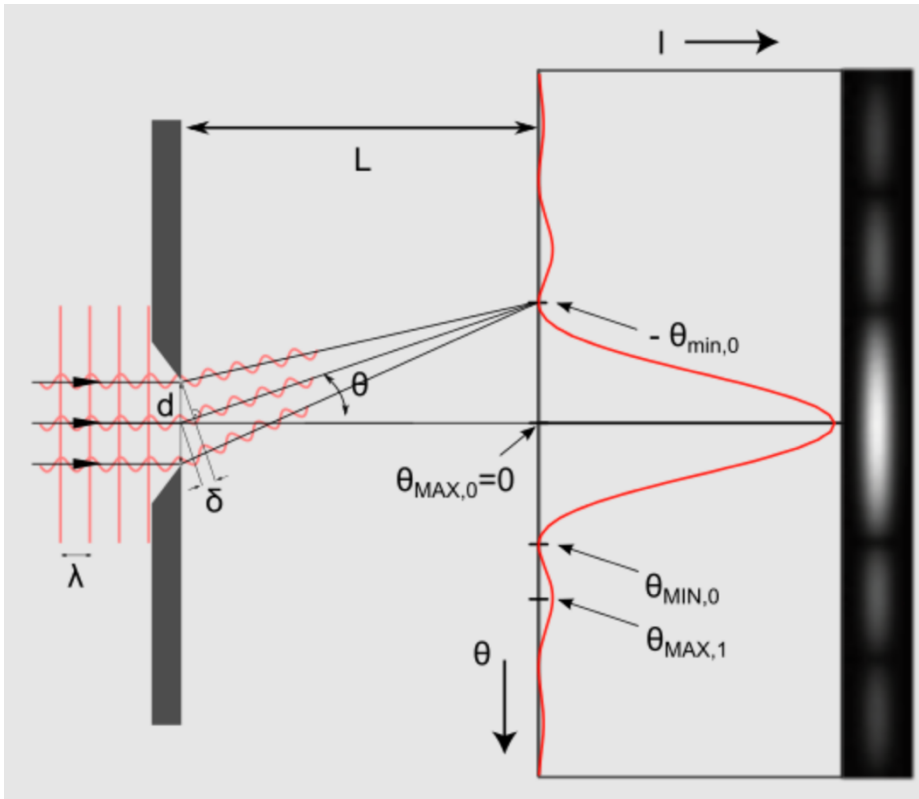
Central frequency (GHz)	Bandwidth (MHz)	Polarisations	NEΔT	IFOV	Pixel
6.925	350	V, H	0.3 K	40x70 km	50x50 km
10.65	100	V, H	0.7 K	26x46 km	50x50 km
18.7	200	V, H	0.7 K	15x26 km	25x25 km
23.8	400	V, H	0.6 K	12x20 km	25x25 km
36.5	1000	V, H	0.7 K	7.6x13 km	15x15 km
50.2	200	V, H	1.8 K	5.5x9.6 km	10x10 km
53.8	400	V, H	1.6 K	5.2x9.0 km	10x10 km
89.0	3000	V, H	1.2 K	3.1x5.4 km	5x5 km

<https://space.oscar.wmo.int/instruments/view/amsr>

IFOV: Instrument field of view



Why do different frequencies have different spatial resolutions?



width depends on

$$\frac{\text{wavelength}}{\text{aperture diameter}} = \frac{\lambda}{d}$$

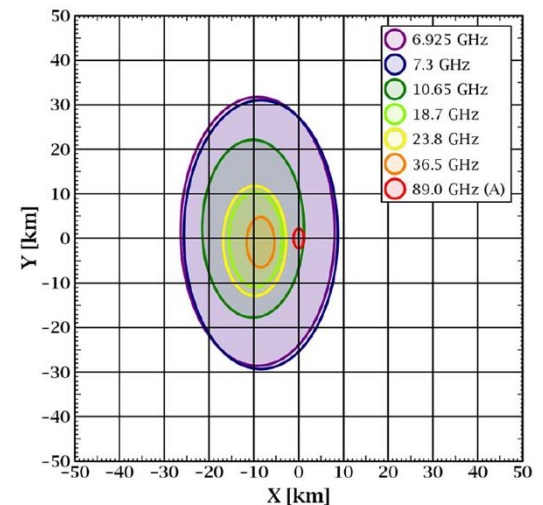
- higher frequencies $\rightarrow$ narrower width

Data file information

Table 6. L1R data and frequencies selected for resampling

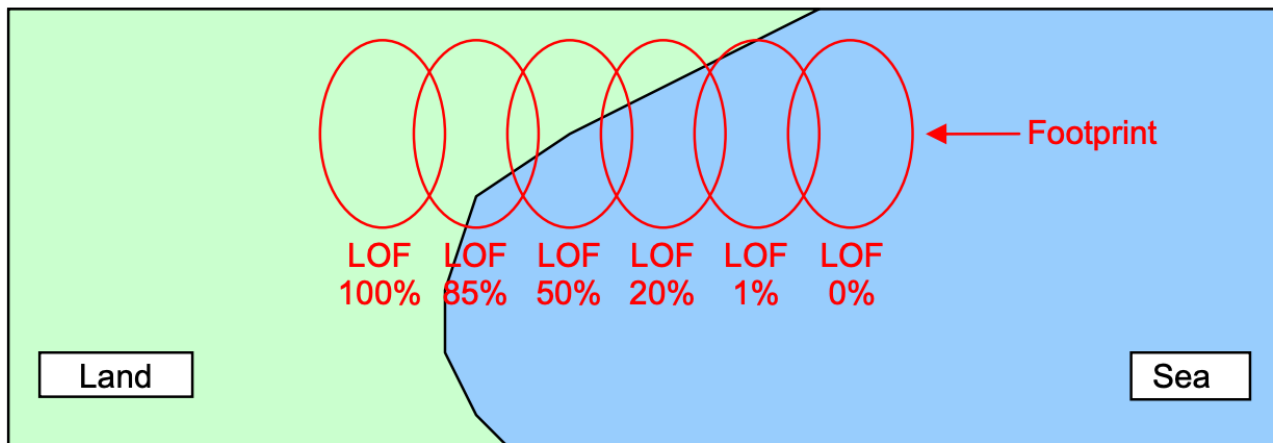
Footprint	6G (H/V) 243 points	7G (H/V) 243 points	10G (H/V) 243 points	18G (H/V) 243 points	23G (H/V) 243 points	36G (H/V) 243 points	89G (H/V) 243 points	89G A-horn (H/V) 486 points	89G B-horn (H/V) 486 points
6GHz	☆	○	○	○	○	○	○		
10GHz			☆	○	○	○	○		
23GHz				○	☆	○	○		
36GHz						☆	○		
89GHz								△	△

- to make the data more useable
the brightness temperatures are
resampled and available at
specific resolutions



Land/Ocean flag

- water pixels that overlap with land are “land contaminated”
- the level of land contamination is measured with a LOF flag
- LOF=100 % - pure land
- LOF=0 % - pure water



Remote sensing products

Levels of satellite products

- Level 0 Reconstructed, unprocessed instrument and payload data at full resolution (eg. sensor counts)
- Level 1 Reconstructed, unprocessed instrument data at full resolution, time-referenced, and annotated with ancillary information, including radiometric and geometric calibration coefficients and georeferencing parameters, Level 1B are Level 1A data that have been processed to sensor units (eg. brightness temperature)
- Level 2 Derived geophysical variables at the same resolution and location as L1 source data (eg. sea ice concentration)
- Level 3 Variables mapped on uniform space-time grid scales, usually with some completeness and consistency (eg., gridded sea ice concentration)
- Level 4 Model output or results from analyses of lower-level data (e.g., variables derived from multiple measurements, such as an ice velocity product that uses sea ice concentration).

paraphrased from:

<https://earthdata.nasa.gov/collaborate/open-data-services-and-software/data-information-policy/data-levels>

Look for documentation!

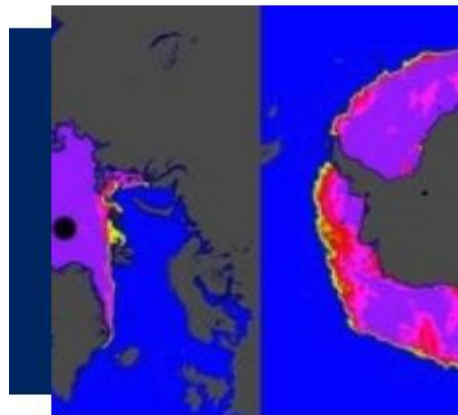
<https://osi-saf.eumetsat.int/products/sea-ice-products>



	NEAR REAL TIME PRODUCT Global Sea Ice Concentration (SSMIS)	OSI-401-b	 Operational	 DMSP/SSMIS	 L3	 1 per day	 5 h	 Global	 10 km
	NEAR REAL TIME PRODUCT Global Sea Ice Edge	OSI-402-d	 Operational	 DMSP/SSMIS and Metop/ASCAT	 L3	 1 per day	 5 h	 global	 10 km
	NEAR REAL TIME PRODUCT Global Sea Ice Type	OSI-403-d	 Operational	 DMSP/SSMIS, Metop/ASCAT and GCOM-W/AMSR-2	 L3	 1 per day	 5 h	 global	 10 km
	NEAR REAL TIME PRODUCT Global Sea Ice Emissivity	OSI-404-a	 Operational	 DMSP/SSMIS	 L3	 1 per day	 5 h	 global	 10 km

Look for documentation!

<https://osi-saf.eumetsat.int/products/sea-ice-products>



Global Sea Ice Concentration (SSMIS)

NEAR REAL TIME PRODUCT

● Operational

Ice concentration is computed from atmospherically corrected SSMIS brightness temperatures, using a combination of state-of-the-art algorithms. The product series is operational since 2005 (version OSI-401-b since 26/05/2016). This product is complementary to the AMSR-2 global sea ice concentration product (OSI-408).

Identification

Acronym :	GBL SICO	Identifier :	OSI-401-b
Product navigator reference :	EO-EUM-DAT-DMSP-OSI-401-B	Acronym for EDC (APNM) :	OSICOGB
OSI SAF producer :	Danish Meteorological Institute		

Characteristics

Processing level :	L3	Spatial coverage :	Global
Satellite input :	DMSP/SSMIS	Spatial sampling :	10 km
Other input :	ECMWF outputs	File Formats :	HL FTP Server (NetCDF3), HL FTP Server (HDF5), HL FTP Server (GRIB1), EUMETCast (GRIB1), EUMETCast (NetCDF3)
Frequency :	1 per day	Characteristics & methods :	Daily averaged fractional ice cover in percentage.
Central time :	12:00		
Timeliness :	5 h		

Data Access

Information

Description →

Quicklook →

Quality →

Production status →

Documentation

PUM →

ATBD →

SVR →

Quality control flags in remote sensing products

- quality flags are usually binary numbers where bits are set or unset indicating if certain flags are set
- For example, for level 2 sea ice concentration (OSI-SAF product)

Bit	Value	Definition
1	1	Position is over land
2	2	Position is over lake
3	4	SIC set to zero by open water filter
4	8	2 m air temperature is high
6	32	Value is the result of spatial interpolation
7	64	Value is the result of temporal interpolation
8	128	SIC is zero, outside of climatological region

On your own: Some ideas

- Plot the brightness temperature for a region of interest for a range of the frequencies in the netcdf file given and for hpol and vpol. Comment on how and why they look different for some of the ocean regions, relating to the atmospheric conditions (winds, cloud water and water vapour), and potential sea ice properties (based on region and season). You may wish to look at ERA5 data at the same time/day/location
- Write a small summary of why horizontal and vertical emissivities are different based on Fresnel's equations. A good reference is Chapter 2 of Ulaby and Long (Microwave Radar and Radiometric Remote Sensing)
- Find more examples of active and passive sensing, either from satellite based systems or other types of systems, and explain a few differences between the two sensing modes for each

Remote sensing references

Natural Resources Canada tutorials:

<https://www.nrcan.gc.ca/maps-tools-and-publications/satellite-imagery-and-air-photos/tutorial-fundamentals-remote-sensing/9309>

Microwave resources from the COMET program:

http://stream1.cmatc.cn/pub/comet/QPF-QPE/microwave/comet/npoess/microwave_topics/resources/index.htm

Textbooks

- Rees, W.G., 2012, Physical Principles of Remote Sensing, Cambridge University Press.
- Tempfli, et al., Principles of remote sensing: an introductory textbook, <https://research.utwente.nl/en/publications/principles-of-remote-sensing-an-introductory-textbook-4>

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References

- Alerskans, E., Hoyer, J.L. Gentemann, C.L., Pedersen, L.T., Nielsen-Englyst, P. and Donlon, C., 2020, Construction of a climate data record of sea surface temperature from passive microwave measurements, *Remote Sensing of Environment*, 236, 111485.
- Brasnett, B. and Surcel Colan, D., 2016, Assimilating Retrievals of Sea Surface Temperature from VIIRS and AMSR2, *Journal of Atmospheric and Oceanic Technology*, 33, 361-375.
- Drucker, R., Martin, S. and Moritz, R.E., 2033, Observations of ice thickness and frazil ice in the St Lawrence Island polynya from satellite imagery, upward looking sonar and salinity/temperature moorings, *Journal of Geophysical Research, Atmospheres*, 108(5), 10.1029/2001JC001213.
- Ivanova et al., 2015, Inter-comparison and evaluation of sea ice algorithms: towards further identification of challenges and optimal approach using passive microwave observations, *The Cryosphere Discussions*, 9, 1797-1817
- Lavergne et al., 2019, Version 2 of the EUMETSAT OSI SAF and ESA CCI sea-ice concentration climate data records, *The Cryosphere*, 13, 49-78, 2019.
- Liu, J. et al., Automatic detection of the ice edge in SAR Imagery Using Curvelet Transform and Active Contour, *Remote Sensing*, 8(6),480, 2016.
- Maeda, T. 2020, Spatial resolution enhancement algorithm based on Backus-Gilbert method and its application to GCOM-W AMSR2 data, *IEEE Transactions on Geoscience and Remote Sensing*, 58(4), 2809- 2816.
- Minnett, P.J. et al., 2019, Half a century of satellite remote sensing of sea-surface temperature, *Remote Sensing of Environment*, 233, 111366.
- Nielsen-Englyst P., Hoyer, J.L., Alerskans, E., Pedersen, L.T. and Donlon, C , 2021, Impact of channel selection on SST retrievals from passive microwave observations, *Remote Sensing of Environment*, 254, 112252.

References

Rees, W.G., 2012, Physical Principles of Remote Sensing, Cambridge University Press.

Reul et al., 2020, Sea surface salinity estimates from spaceborne L-band radiometers: An overview of the first decade of observation (2010-2019), Remote Sensing of Environment, 242, 111769.

Scott et al., 2012, Direct assimilation of AMSR-E brightness temperatures for estimating sea ice concentration, Monthly Weather Review, 140, 997-1013.

Shokr and Sinha, 2015, Sea ice physics and remote sensing, Geophysical Monograph 209, American Geophysical Union and John Wiley & Sons, Inc.

Strong, C. and Rigor, I.G., 2013, Arctic marginal ice zone trending wider in summer and narrower in winter, Geophysical Research Letters, 40, 4864-4868.

Tempfli et al. (eds), Principles of Remote Sensing, An introductory textbook, The International Institute for Geo-Information Science and Earth Observation, 2009

Ulaby, F.T. and Long, D.G., Microwave radar and radiometric remote sensing, The University of Michigan Press, 2014.

Wentz, F.J. and Meissner, T., (2007) Supplement 1, Algorithm Theoretical Basis Document for AMSR-E Ocean Algorithms, RSS Technical Report, 051707.

Wilheit, T. T., 1979, A model for the microwave emissivity of the ocean's surface as a function of wind speed, IEEE Transactions on Geoscience and Electronics, 17, 244–249.